UNIT 5 SERVICE LAYER PROTOCOLS & SECURITY

IoT Service Layer Protocols & Security Protocols:

It can be categorized based on their functionality and purpose.

IoT Service Layer Protocols: It refers to a set of communication protocols and standards designed to facilitate interaction and data exchange between various IoT devices, applications, and services. These protocols operate at the service layer of the IoT architecture and play a crucial role in enabling seamless communication, interoperability, and management of IoT devices and data.

Purpose and Functions:

1. Interoperability:

Service layer protocols aim to ensure interoperability among diverse IoT devices and applications. They provide a standardized way for different devices, often from various manufacturers, to communicate and work together.

2. Abstraction of Complexity:

These protocols abstract the underlying complexity of the IoT network and device communication, providing a common interface and set of functionalities that developers can leverage.

3. Common Service Abstractions:

Service layer protocols define common service abstractions, such as data management, device discovery, security mechanisms, and application enablement. This allows developers to create applications without needing to understand the intricacies of each device's communication protocol.

Common IoT Service Layer Protocols:

1.MQTT (Message Queuing Telemetry Transport):

• Lightweight and efficient messaging protocol suitable for low-bandwidth, high-latency, or unreliable networks.

• Commonly used for real-time communication between IoT devices and applications, especially in scenarios where low power and low data overhead are crucial.

2. CoAP (Constrained Application Protocol):

• Designed for resource-constrained devices and networks, providing a lightweight alternative to traditional HTTP for communication in IoT.

● Well-suited for applications where devices may have limited resources, such as in industrial IoT or sensor networks.

3. oneM2M (One Machine-to-Machine):

● A global standard that aims to provide a common service layer for M2M and IoT applications, ensuring interoperability across diverse platforms.

● Enables seamless communication and data exchange between devices and applications, promoting a standardized approach to IoT deployments.

4. OMA (Open Mobile Alliance) Standards:

● OMA develops various standards for mobile services, including those relevant to IoT. Examples include Lightweight M2M (LwM2M) for device management and Smart Objects for defining data models.

● Addresses specific needs in IoT ecosystems, offering solutions for device management, data representation, and service enablement.

Characteristics:

1. Scalability:

The protocols are designed to scale efficiently, supporting communication between a large number of devices and applications within the IoT ecosystem. **2. Security:** Security features are often embedded in these protocols, addressing authentication, encryption, and secure data transfer to protect IoT devices and data from unauthorized access or tampering.

3. Flexibility:

Protocols are flexible to accommodate different types of IoT devices, communication patterns, and application requirements.

4. Standards Compliance:

Attachment to industry standards ensures that devices and applications from different vendors can work together seamlessly, fostering interoperability. IoT Service Layer Protocols form the foundation for effective communication and collaboration in IoT ecosystems, providing a standardized and efficient devices and applications.

Service Layer in oneM2M:

The **service layer** in oneM2M is responsible for providing services to applications, end devices, and other entities in the system. It operates on top of the **oneM2M architecture**, ensuring that the necessary functionality (like data management, security, and communication) is abstracted and exposed to the application layer.

Key functions of the **service layer** include:

1. **Abstraction of Device/Thing Management:**
    - It abstracts the underlying hardware and device management details, allowing the application to interact with the devices without worrying about the specifics of hardware or communication protocols.
2. **Application Services:**
    - The service layer offers different services that can be used by applications (like data storage, event monitoring, communication management) in a standardized way. This simplifies the development of IoT applications by providing common services across devices and platforms.
3. **Interoperability:**
    - Through the service layer, oneM2M ensures interoperability between different devices, networks, and protocols. It standardizes how devices and applications should communicate with each other, overcoming the challenges of working with heterogeneous IoT systems.
4. **Security and Access Control:**
    - The service layer includes security mechanisms like authentication, encryption, and access control policies. This ensures that only authorized devices or applications can access the IoT resources and data.
5. **Resource Management:**
    - The service layer is responsible for managing the resources (like data, applications, services) that are offered in the system. This includes handling service discovery, allocation, and updates.

ETSI M2M architecture:

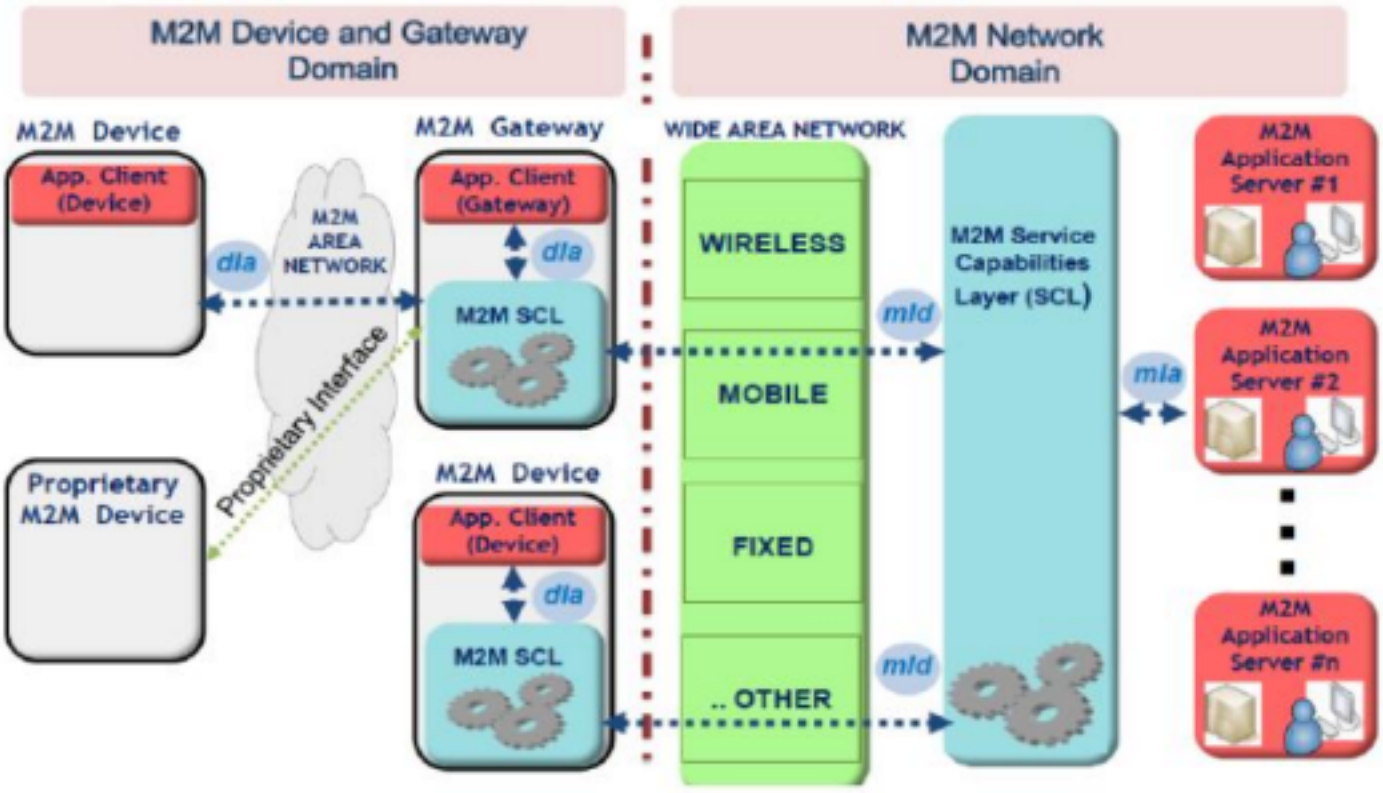


Fig: M2M Functional Architecture Overview

Fig: M2M Architectural framework

M2M Service Capabilities Layer provides functions that are to be exposed on the reference points. M2M SCs can use Core Network functionalities through a set of exposed interfaces (e.g. existing interfaces specified by 3GPP, 3GPP2, ETSI TISPAN, etc.). Additionally, M2M SCs can interface to one or several Core Networks.

The following terms will be used are :

• **NSCL**: Network Service Capabilities Layer refers to M2M Service Capabilities in the Network Domain.

• **GSCL:** Gateway Service Capabilities Layer refers to M2M Service Capabilities in the M2M Gateway.
• **DSCL:** Device Service Capabilities Layer refers to M2M Service Capabilities in the M2M Device.
• **SCL:** Service Capabilities Layer refers to any of the following: NSCL, GSCL, or DSCL. ● **D/G SCL:** refers to any of the following: DSCL, GSCL.
The external reference points (mIa, mId, dIa) are mandated and are required for ETSI M2M compliance.
M2M Node: is a logical representation of the M2M components in the M2M Device, M2M Gateway, or the M2M Core. An M2M Node shall include one SCL, and optionally an M2M Service Bootstrap function and an M2M Service Connection function. An M2M Node relies on a Secured Environment Domain, controlled by the M2M Service Provider associated with the SCL, to protect Sensitive Functions and Sensitive Data. A Device/Gateway M2M Node shall be represent upon pre-provisioning or executing an M2M Service Bootstrap procedure on the M2M Device/Gateway with an M2M Service Provider. Each Device/Gateway M2M Node may be represent with only one M2M Service Provider.
M2M Applications: are respectively Device Application (DA), Gateway Application (GA) and Network Application (NA). DA could reside in an M2M Device which implements M2M Service Capabilities or alternatively reside in an M2M Device which does not implement M2M Service.

Open Mobile Alliance (OMA):

The **OMA** is an industry that develops open standards for mobile services, applications, and platforms. Founded in 2002, OMA's goal is to encourage the development of global standards for mobile and wireless technologies, ensuring interoperability across different networks, devices, and platforms. It plays a pivotal role in shaping the future of mobile communications, content delivery, and mobile applications.

Objectives of the Open Mobile Alliance:

1. **Interoperability**: One of OMA’s core missions is to enable interoperability between devices, networks, and services. This is crucial in a mobile ecosystem where users might use a variety of devices and be on different networks or service platforms.
2. **Global Standards**: OMA aims to define open and global standards for the mobile industry. These standards help reduce the complexity and cost of mobile service deployment and ensure that applications, content, and services work seamlessly across different environments.
3. **Innovation and Advancement**: OMA actively supports the development of innovative technologies for mobile applications and services. The standards it creates support everything from mobile content delivery to device management, messaging, and more.
4. **Collaborative Development**: The OMA brings together industry stakeholders, including mobile network operators, device manufacturers, content providers, and software developers. By creating a collaborative platform, OMA ensures that a wide range of perspectives and expertise inform its standards.

OMA's Areas of Focus:

OMA covers a wide range of areas in the mobile industry, addressing key aspects of mobile and wireless communications. Some of the core areas include:

1. Mobile Content Delivery

- OMA defines standards for delivering mobile content, such as multimedia, images, and text, over cellular networks. These standards help ensure that content is delivered efficiently and can be consumed on a wide variety of devices.
- **This** Standards like OMA DRM (Digital Rights Management) ensure that content is protected and can be securely accessed and shared.

2. Device Management

- Device management is a critical function in mobile networks, particularly in large-scale deployments such as enterprise devices and IoT (Internet of Things) applications.
- This standard allows service providers and enterprises to remotely manage devices, update software, configure settings, and ensure devices are functioning properly without physical interaction.

3. Messaging Services

- OMA is also involved in developing standards for messaging, including traditional SMS (Short Message Service) and more advanced messaging types. • Defines services for voice communication, allowing instant communication between users, similar to a walkie-talkie.
- Supports messaging and presence functionality across mobile platforms, allowing users to see each other's status and communicate in real time.

4. Location-Based Services (LBS)

- OMA helps define standards for location-based services, allowing mobile applications to provide services based on the geographic location of the user. This is key for many modern mobile applications like ride-hailing, navigation, and location-based advertising.

5. Network Services and Service Delivery

- OMA defines how services should be delivered over mobile networks, ensuring that they are compatible with the underlying infrastructure. This includes defining protocols for service discovery, management, and interaction.
- A framework for enabling service providers to deliver multimedia and other services over mobile networks.
- Exposes network and service provider capabilities to third-party developers via APIs, making it easier for third-party apps to access network services.

6. Security Standards

- Security is a critical aspect of mobile services, particularly with the growing use of mobile devices for financial transactions, communications, and IoT applications.

- OMA creates standards for secure communication, device authentication, and privacy protection to ensure that users can trust the services they are using.

7. Mobile Broadband and IoT

- OMA's standards also support the evolution of mobile broadband technologies and the IoT. This includes defining the requirements for high-speed data services, device-to-device communication, and the management of IoT networks and devices.

Standards and Specifications by OMA:

- **OMA Device Management (DM)**: Defines how mobile devices can be remotely managed, including software updates and configuration changes.
- **OMA Digital Rights Management (DRM)**: Specifies how digital content is protected from unauthorized access, ensuring that content providers can monetize their digital assets securely.
- **OMA Service Delivery Platform (SDP)**: A framework that supports the delivery of content and services across mobile networks.
- **OMA Push-to-Talk (PTT)**: Standardizes the way voice communication is handled over mobile networks, providing a "walkie-talkie" style communication service.
- **OMA Instant Messaging and Presence Service (IMPS)**: Defines a framework for instant messaging and presence awareness on mobile networks. • **OMA Mobile Location Protocol (MLP)**: Provides a standard for location-based services (LBS) to share the geographic position of a device.
- **OMA Mobile Web (MWB)**: Standards for providing web-based content and services to mobile devices.
- **OMA Smart Home**: The development of standards that enable devices and applications within the home to be interconnected, forming the basis of smart home technologies.

OMA's Role in the IoT:

The Open Mobile Alliance plays a key role in defining standards for the Internet of Things (IoT). As IoT grows, OMA helps to define the protocols and frameworks that ensure the smooth interaction of IoT devices across networks. Some of the IoT-related activities OMA is involved in include:

- **OMA Lightweight M2M (LwM2M)**: A protocol for managing IoT devices, ensuring low power consumption, and supporting remote management and communication between devices.
- **OMA Smart Appliances**: Defining standards for interconnected devices within a smart home, such as thermostats, lighting, and appliances.

The Broadband Forum (BBF):

The Broadband Forum (BBF) is a combination of approximately 200member companies that work collaboratively to define broadband network specifications, management systems, and various related technologies. Initially, the focus of BBF was on broadband access technologies, but its scope has expanded to encompass multiple areas including home networking, Internet of Things (IoT), cloud services, and more.

BBF and IoT:

Within the realm of the Internet of Things (IoT), the Broadband Forum has contributed in several ways:

1.TR-069 Protocol:

• BBF developed the TR-069 protocol, also known as the CPE WAN Management Protocol (CWMP). While initially designed for remote management of Customer Premises Equipment (CPE) by Internet Service Providers (ISPs), it has extended its functionality to manage IoT devices as well.

• TR-069 offers a standardized and secure approach for remote management, provisioning, and configuration of IoT devices, ensuring interoperability across various devices and networks.

2. Interoperability and Standards:

• BBF has been involved in establishing standards and specifications that promote interoperability among different IoT devices and platforms.

• It aims to create a standardized framework for IoT deployments, enabling seamless communication and compatibility between devices from various manufacturers. **3. Home Automation and IoT Security:**

• With a focus on home networking and connected devices, BBF works on defining standards for home automation and IoT security within broadband environments.

• This includes developing best practices and protocols to ensure secure communication and data protection in IoT devices connected via broadband networks.

4. Collaboration and Industry Engagement:

• BBF collaborates with other standardization bodies and industry groups to ensure alignment and convergence of IoT-related standards and practices.

• By engaging with various stakeholders, BBF aims to drive innovation and development of IoT technologies while emphasizing security and interoperability.

IoT Security Protocols:

IoT Security Protocols are a set of protocols and standards designed to secure the communication, data, and interactions within IoT ecosystems. Rapid growth of IoT devices and the potential vulnerabilities associated with their deployment, ensuring robust security is essential to protect against unauthorized access, data breaches, and various cyber threats.

Common IoT Security Protocols:

1. TLS (Transport Layer Security) and DTLS (Datagram Transport Layer Security): • **It** Provide secure communication channels by encrypting data exchanged between IoTdevices and servers.

● Ensures confidentiality and integrity of data, preventing eavesdropping and tampering during transmission.

2. OAuth (Open Authorization):

• **It** Enables secure authorization between IoT devices and services without sharing sensitive credentials.

• Allows devices to access resources or services on behalf of a user with proper authentication, enhancing security and user privacy.

3. X.509 Certificates:

● Defines a standard format for public key certificates, supporting authentication in IoT environments.

● Certificates are used to verify the identity of devices and establish secure communication channels, often employed in conjunction with TLS.

4. CoAP (Constrained Application Protocol) with DTLS/TLS:

● **It** Extends the CoAP protocol with DTLS/TLS for secure communication in resource constrained IoT devices.

● Ensures secure data exchange in environments where devices have limited processing power and memory.

5. JWT (JSON Web Tokens):

● **It** Provides a compact, URL-safe means of representing claims to be transferred between parties.
● Often used for secure authentication and authorization, especially in scenarios where lightweight token-based authentication is preferred.
6. LwM2M (Lightweight M2M) Security:
● **It** includes security mechanisms for device management in IoT, addressing issues like authentication and secure firmware updates.
● Ensures the secure management of IoT devices, protecting against unauthorized access and malicious activities.

Security Characteristics:

1. Authentication: Ensures that devices and users are who they claim to be, preventing unauthorized access.
2. Authorization: Controls access to resources, ensuring that only authorized entities can perform specific actions or access certain data.
3. Encryption: Protects data in transit by encrypting it, preventing securely and unauthorized interception.
4. Integrity: Ensures the integrity of data, preventing unauthorized tampering or modification.
5. Secure Bootstrapping and Provisioning: Establishes a secure initial setup for devices, ensuring they are configured and authenticated securely from the start.
6. Secure Device Lifecycle Management: Addresses security considerations throughout the entire lifecycle of an IoT device, including provisioning, operation, and decommissioning. **7. Security Standards Compliance:** Established security standards and best practices, ensuring a consistent and well-understood security posture for IoT implementations.

MAC 802.15.4:

IEEE 802.15.4 is a standard that defines the physical (PHY) and medium access control (MAC) layers for low-rate wireless personal area networks (LR-WPANs). It is designed to provide a low-power, low-cost communication solution with a limited range, typically between 10 and 100 meters.

The main features of IEEE 802.15.4 include:

- **Low Power Consumption:** It is designed for battery-operated devices, with a focus on long battery life.
- **Low Data Rates:** The standard supports data rates of 250 kbps, 100 kbps, and 20 kbps depending on the frequency band used.
- **Communication Range:** Typically up to 100 meters in open space. • **Frequency Bands:** IEEE 802.15.4 operates in the 2.4 GHz, 868 MHz, and 915 MHz ISM bands, depending on the region.
- **Simple MAC Protocol:** The MAC protocol is simple and supports low-latency communication, which is ideal for small data transmissions in a wireless sensor network.
- **Supports Star and Peer-to-Peer Topologies:** It can operate in star, peer-to peer, and mesh network configurations.

Key components of the MAC layer include:

- **Frame Structure:** IEEE 802.15.4 uses a frame structure to handle data

transmission, including data, acknowledgment, and command frames. • **Channel Access Methods:** It uses a contention-based access method CSMA/CA and an optional deterministic access method for certain types of networks.
• **Security:** It includes basic security features like encryption (AES) and authentication.

6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks): It is a standard designed to enable the use of the Internet Protocol (IPv6) on constrained, low-power, wireless devices that typically communicate over short-range networks. It is primarily used in the context of **Internet of Things (IoT)** applications, where devices are constrained in terms of power, processing capabilities, and memory. 6LoWPAN operates over IEEE **802.15.4** networks, which provide a simple, low-power, and low-data-rate communication standard.

Features of 6LoWPAN:

1. **IPv6 Adaptation**:
    - **6LoWPAN** enables the transmission of **IPv6 packets** over low-power wireless networks, such as those based on the IEEE 802.15.4 standard, which is commonly used in IoT.
    - It solves the problem of using IPv6 in environments with **limited bandwidth**, **small packet sizes**, and **low processing power** by adapting the large IPv6 headers and protocols to fit these constraints. 2. **Compression of IPv6 Headers**:
    - One of the primary challenges in using IPv6 over constrained networks is the size of the IPv6 headers, which are too large for the small packet sizes that IEEE 802.15.4 supports.
    - **6LoWPAN** uses **header compression** to reduce the size of the IPv6 packet headers. This reduces overhead, making the network more efficient.
    - For example, a typical IPv6 header is 40 bytes, but 6LoWPAN compresses the headers down to just a few bytes, making it possible to transmit IPv6 packets within the 802.15.4 frame size of 127 bytes.
3. **Fragmentation and Reassembly**:
    - 6LoWPAN allows **fragmentation** of IPv6 packets if they are larger than the maximum transmission unit (MTU) supported by IEEE 802.15.4, which is 127 bytes.
    - The 6LoWPAN protocol handles breaking down large IPv6 packets into smaller fragments on the sender side and **reassembling** them on the receiver side.
4. **Efficient Routing**:
    - **6LoWPAN** provides a simple, efficient way of routing IPv6 packets in constrained environments. It supports routing techniques suitable for small and low-power devices.
    - The use of **neighbor discovery** and **stateless address autoconfiguration (SLAAC)** protocols within 6LoWPAN enables devices to join the network and communicate without the need for manual configuration.
5. **Mesh Networking Support**:

- **6LoWPAN** allows devices to form a **mesh network**, which improves reliability and range by enabling devices to communicate through other devices rather than requiring direct communication with a central coordinator.
- This is especially useful in **large-scale IoT deployments** where devices may be far apart or in difficult-to-reach locations.

6. **Security**:
   - While **6LoWPAN** provides the foundation for secure communication through encryption mechanisms like **IPsec** (Internet Protocol Security) and **DTLS** (Datagram Transport Layer Security), it is the responsibility of higher layers (such as those used in protocols like Zigbee or Thread) to implement more comprehensive security protocols.
   - For instance, **IPsec** can be used at the IPv6 layer to secure the communication, and **AES-128** encryption is commonly employed for confidentiality in IoT networks.

Applications of 6LoWPAN:

6LoWPAN is well-suited for **low-power IoT devices** that need to be connected to the Internet. Some of the main applications include:

1. **Smart Home**:
   - Devices like thermostats, smart lighting, and door sensors can use 6LoWPAN to communicate with each other and central controllers over the internet.
2. **Industrial IoT**:
   - In industrial settings, 6LoWPAN enables sensors and actuators to communicate within factory automation systems or process monitoring systems.
3. **Healthcare**:
   - Medical devices like wearable sensors for monitoring heart rate, blood pressure, or other vital signs can use 6LoWPAN to transmit data to a health monitoring system.
4. **Agriculture**:
   - Smart farming applications can use 6LoWPAN for collecting data from sensors in fields, such as temperature, moisture levels, or soil conditions, and transmit the data to cloud-based systems for analysis.
5. **Smart Cities**:
   - 6LoWPAN can enable applications like smart street lighting, waste management, and environmental monitoring, where devices need to communicate over long distances with low power consumption.

Security in IoT Protocols – RPL (Routing Protocol for Low Power and Lossy Networks):

RPL is a routing protocol designed for **low-power**, **lossy networks (LLNs)**, which are commonly used in **Internet of Things (IoT)** applications. It is part of the **IPv6** suite, defined in **RFC 6550**, and is typically employed in environments where devices have limited resources (in terms of power, bandwidth, and computational capacity), such as **smart homes**, **healthcare**, and **industrial applications**. RPL is particularly

useful in **6LoWPAN** networks, enabling IPv6 communication over low-power wireless networks (such as IEEE 802.15.4).

RPL enables efficient routing in these constrained networks by forming a **Directed Acyclic Graph (DAG)** for data transmission. Although RPL is designed to be lightweight and efficient, security is a significant concern due to the nature of IoT networks, which are vulnerable to various attacks like **eavesdropping**, **manipulation**, and **DoS (Denial of Service)**.

Security Requirements in RPL: In the context of **IoT networks**, security is essential to ensure the **confidentiality**, **integrity**, and **availability** of data. RPL networks must protect against a range of potential attacks that could compromise network communication. Here are the key security goals and challenges for RPL:

1. **Confidentiality**: Ensure that sensitive data transmitted between devices is not exposed to unauthorized parties.
2. **Authentication**: Ensure that only authorized devices can join the network and participate in routing.
3. **Integrity**: Ensure that routing information and data packets are not tampered with during transmission.
4. **Availability**: Ensure that the network can withstand attacks such as DoS and maintain service even under attack.
5. **Non-repudiation**: Ensure that devices cannot deny their actions, and that communication can be traced back to the origin.

RPL Security Mechanisms

RPL incorporates a variety of security mechanisms to address these requirements: 1. **Authentication of RPL Control Messages**:

- The integrity of **DIO (DODAG Information Object)** and **DAO (Destination Advertisement Object)** control messages in RPL must be ensured to prevent malicious manipulation.
- RPL employs cryptographic techniques like **Message Authentication Code (MAC)** or **digital signatures** to authenticate control messages and ensure they are from legitimate nodes.

2. **Confidentiality and Encryption**:
   - While RPL itself doesn't provide encryption, it can be used with other security protocols such as **IPsec** (for IP layer security) or **DTLS** (Datagram Transport Layer Security) to ensure the confidentiality and integrity of data transmitted between nodes.
   - End-to-end encryption can protect sensitive IoT data, such as health information or industrial control data, from eavesdropping.
3. **Protection Against Routing Attacks**:
   - **Replay Attacks**: To protect against replay attacks, RPL employs **sequence numbers** in control messages to ensure the freshness of data and prevent an attacker from reusing old messages.
   - **Blackhole Attacks**: In blackhole attacks, a malicious node might drop or discard packets. RPL combats this by using **neighbor discovery** and **secure message validation** to detect and isolate malicious nodes.
   - **Sybil Attacks**: A Sybil attack occurs when an attacker presents multiple fake identities to manipulate the network. RPL addresses this by

implementing **device authentication** and using cryptographic techniques to verify node identities.

4. **Route Validation and Integrity**:
   - RPL ensures that the route information passed between nodes is not tampered with by using **cryptographic hashes** and **digital signatures**. This prevents attackers from injecting false routing information or altering the structure of the DAG.
5. **Neighbor Discovery and Secure Routing**:
   - **Secure Neighbor Discovery (SEND)** can be integrated with RPL to protect against malicious neighbors pretending to be legitimate nodes.
   - Authentication protocols are used to verify that a node wishing to join the network is authorized, and unauthorized nodes can be excluded.
6. **Message Integrity with Cryptographic Hashing**:
   - RPL uses **hash functions** to ensure the integrity of control messages. For example, each **DAO message** can include a hash of the routing data, which is then validated by the receiving node to ensure that no tampering has occurred.